ON IMAGE TRANSMISSION USING COOPERATIVE NOMA IN VLC NETWORK

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UNDER THE GUIDANCE OF

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DECLARATION BY CANDIDATES

We hereby declare that the Report of the Project Work entitled "**ON IMAGE TRANSMISSION USING COOPERATIVE NOMA IN VLC NETWORK**" which is being submitted to the **Electronics & Communication Engineering Department, MANIT BHOPAL**, in the partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering** in **the Department of Electronics & Communication Engineering**, is a bonafied report of the work carried out by us. The material contained in this report has not been submitted to any University or Institution for the award of any degree.

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CERTIFICATE OF APPROVAL

It is certified that the work contained in this report titled, "ON IMAGE TRANSMISSION USING COOPERATIVE NOMA IN VLC NETWORK", is the original work done by the students and has been carried out under my supervision.

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ABSTRACT

This project presents a framework to enhance image transmission in Visible Light Communication (VLC) systems by integrating Cooperative Non-Orthogonal Multiple Access (NOMA) techniques. VLC, a promising alternative to RF communication, offers advantages such as high bandwidth and immunity to electromagnetic interference. However, image transmission over VLC faces challenges like limited link reliability and dynamic variation.

To address these issues, this work explores the potential of cooperative NOMA, where strong users assist weak users in decoding signals, thereby improving system throughput and reliability. An image is encoded, modulated, and transmitted using VLC with NOMA-enabled power allocation. A deep learning-based decoder is used to reconstruct image.

Simulation results show significant improvements in image quality (PSNR) and BER under different channel conditions compared to conventional schemes. This cooperative VLC-NOMA approach holds promise for future smart lighting and indoor multimedia applications.

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TABLE OF CONTENTS

Declaration by the candidate ..... **i**

Abstract ..... **ii**

Acknowledgement ..... **iii**

Table of contents ..... **iv**

List of Figures ..... **vi**

List of Tables ..... **vii**

List of Symbols and Abbreviations ..... **viii**

1. Introduction **1**

1.1 Role of Cooperative Noma 1

1.2 Objective of Study 1

1.3 Background 1

1.4 Motivation and Problem Statement 2

2. DataSet, Materials and Methods **3**

2.1 Related Work on VLC 3

2.2 Signal Structure 4

2.3 Performance Parameters 5

3. Experiments, Results **7**

3.1 Performance Parameters Plots 7

3.2 Output Images 8

3.3 Parameters 9

4. Applications **10**

4.1 Indoor Image Transmission 10

4.2 Smart Indoor Surveillance 10

4.3 Industrial Automation & Monitoring 11

4.4 Healthcare & Medical Imaging 11

5. Conclusion and Future Work 12

6. References 13

7. Publication 14

8. Research Paper 15

LIST OF FIGURES

Name of Figure **Page No.**

2.1 System Model for 2 User Cooperative VLC System…….....4

2.2 Enhanced Image Transmission……5

3.1(a)MSE VS SNR……...7

3.1(b)PSNR VS SNR……..7

3.1(c)NCC VS SNR……7

3.1(d)NAE VS SNR……7

3.1(e)LMSE VS SNR……..8

3.1(f)BER VS SNR…….8

3.2(a)Input image1……..8

3.2(b)Input image2……..8

3.2(c)Output image1……8

3.3(d)output image2 ……8

LIST OF TABLES

Name of Tables **Page No.**

3.3 Performance Parameters……..9

LIST OF SYMBOLS AND ABBREVIATIONS

Symbol / Abbrev.	Description
A	Physical Area of the PD
BER	Bit Error Rate
DF	Decode and Forward
FoV	Field of View
H_LOS	Line-of-Sight Channel Gain
LED	Light Emitting Diode
LMSE	Laplacian Mean Squared Error
LOS	Line of Sight
m	Lambertian Emission Order
MRC	Maximum Ratio Combining
MSE	Mean Square Error
NAE	Normalized Absolute Error
NCC	Normalized Cross Correlation
NOMA	Non-Orthogonal Multiple Access
OOK	On-Off Keying
P_avg	Average Transmitted Power
PD	Photodetector
PSNR	Peak Signal-to-Noise Ratio
R	Responsivity of the Detector
RF	Radio Frequency
SNR	Signal-to-Noise Ratio
VLC	Visible Light Communication
η	Additive White Gaussian Noise
Φ_c	Field of View of PD
ψ	Irradiance Angle

1. INTRODUCTION

1.1 ROLE OF COOPERATIVE NOMA IN VLC

- Non-Orthogonal Multiple Access (NOMA) allows multiple users to share the same channel using power-domain multiplexing.
- Integrating Cooperative NOMA with VLC systems can:
    - Extend coverage
    - Improve spectral efficiency
    - Enhance image transmission quality

1.2 OBJECTIVE OF THE STUDY

- Enhancing performance of image data transmission over Cooperative VLC Network
- Evaluate system performance using:
    - PSNR (Peak Signal-to-Noise Ratio)
    - BER (Bit Error Rate)
    - MSE(Mean Square Error)
    - NCC(Normalized Cross Correlation)
    - NAE(Normalized Absolute Error)
    - LMSE(Laplacian Mean Squared Error)
- Compare performance with conventional VLC systems.

1.3 BACKGROUND

- Traditional RF-based communication systems face issues like limited bandwidth, interference, and security
- Visible Light Communication (VLC) uses LED light for data transmission, offering benefits such as:
    - Large unlicensed bandwidth
    - Reduced electromagnetic interference
    - Higher security due to confined signal propagation
- VLC is well-suited for indoor environments but suffers from:
    - Limited coverage area
    - Strong dependence on line-of-sight (LOS)
    - Susceptibility to shadowing and obstacles

1.4 MOTIVATION AND PROBLEM STATEMENT

- High-quality image transmission needs high data rates and reliable communication.

- Basic VLC systems often experience:
    - High bit error rates
    - Low robustness in obstructed or dynamic environments
- Motivation for this work:
    - Improve image quality (e.g., PSNR) in VLC-based systems
    - Use cooperative strategies to enhance weak user performance
    - Optimize resource allocation in multi-user VLC scenarios

2. DATASET, MATERIALS AND METHODS

2.1 SYSTEM AND CHANNEL MODEL

In particular, consideration is taken for an indoor VLC system with one transmitting and two user nodes (one serving as a relay and the other as a destination). There is a single LED on the transmission node, one LED and one PD on the relay node, and one PD on the destination node. The information is disseminated by the transmitting node, which is located at 4 × 4 × 4m, i.e., in the centre of an 8 × 8 × 4m room's ceiling. With a semi-angle (ψ1/2) source, the Lambertian pattern is taken into account. The receiving portion and the transmission part are the two main components of the relay (U1). A PD is included in the relay's receiving section, and it is evaluated based on its field of view and responsivity (R). The receiving part's primary responsibility is to interpret the message signal and forward it for further transmission. A light source with a semi-angle (ψ1/2) Lambertian pattern and optical power (P tn) are included in the transmitting portion. Sending the message to the primary recipient or another relay is the transmitting part's major responsibility . It has been demonstrated that cooperative transmission is important for VLC systems in situations

where the user occupies a wide area or is not in the field-of-view (FoV). This is handled by taking into account a particular indoor VLC embarking in which the intermediary users function as a secure communication link. Fig. 2.1 depicts the system paradigm for the cooperative OOK modulated VLC system with two users, U1 and U2. There are two phases, i.e., the broadcast phase and the cooperation phase for every bit period. During the broadcast phase, node T x uses OOK modulation to broadcast all user messages. During the cooperative phase, U1 helps U2 transmit a signal so that U2 can receive two signals, one from T x and one from U1. The MRC combining techniques are specifically employed to integrate the signals at the U2

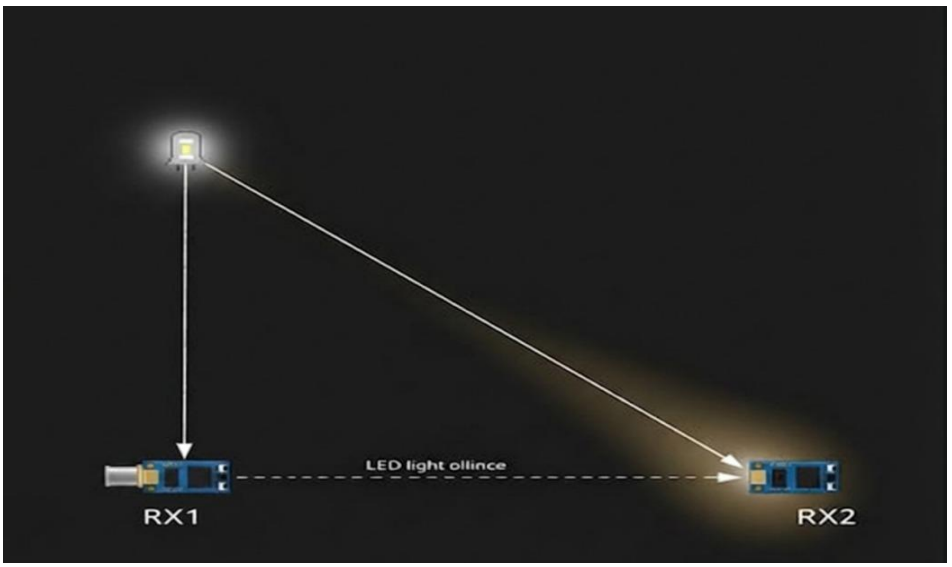


Fig 2.1: System Model for 2 User Cooperative VLC System

The line of sight (LOS) channel impulse response (CIR) is given by :-

$$H_o = \begin{cases} \frac{A}{D_{i,j}^2} R_L(\psi_i)\, G_c(\phi_{i,j})\, G_\zeta(\phi_{i,j}) cos(\phi_{i,j}), for\ 0 \le \phi \le \Phi_c \\ 0, \qquad for\ \phi > \Phi_c \end{cases} \quad ..(1)$$

where, A is physical area of the PD, Di,j is the Euclidean distance between ith transmitter and jth receiver, ϕ is the angle of incidence, Φc denotes the FoV of PD, ψ is the irradiance angle, Gϵ(ϕi,j) is the gain of optical filter, Gζ (ϕi,j) = n2 / sin2(Φc) , for 0 ≤ ϕ ≤ Φc, denotes gain of optical concentrator, n is the refractive index of the optical concentrator, Rι(ψi) = (m+1/ 2π)cosm(ψi,j) is the Lambertian radiant intensity of the ith transmitter and m = − ln(2) / ln(cos(ψ1/2)) is the order of Lambertian emission represented by semi-angle (ψ1/2) at half-power of an LED .

2.2 SIGNAL STRUCTURE

The equation x = 2Pavg d, where d is the sent symbol and Pavg is the average transmitted power, represents the signal sent out at any given moment by the ith transmitting node. The signal received at the jth receiving node as of the ith sending node is provided by,

$$Yj = 2RHLOS\ Pavg\ d + \eta \qquad \ldots. (2)$$

where, HLOS is the LOS channel gain between ith transmitting node and jth receiving node, R is the responsivity of the detector and η is additive white gaussian noise given by

$$\sigma 2\ \eta = \sigma 2\ T + \sigma 2\ S\ , \qquad \ldots (3)$$

where σ2 T and σ2 S are the variances of thermal and shot noise, respectively, and are further defined in and the references therein. At the U1, threshold detection is carried out. Following the DF protocol, the detected signal at U1 is resent to the intended recipient, U2. Two signals, one from the transmitter and one from U1, are received at U2. The two signals are combined using the MRC approach. The weighted signals are combined at the destination by adding them all up. Every signal has its corresponding weight applied to it. As a result, the system's total signal-to-noise ratio (SNR) rises and reception quality improves .

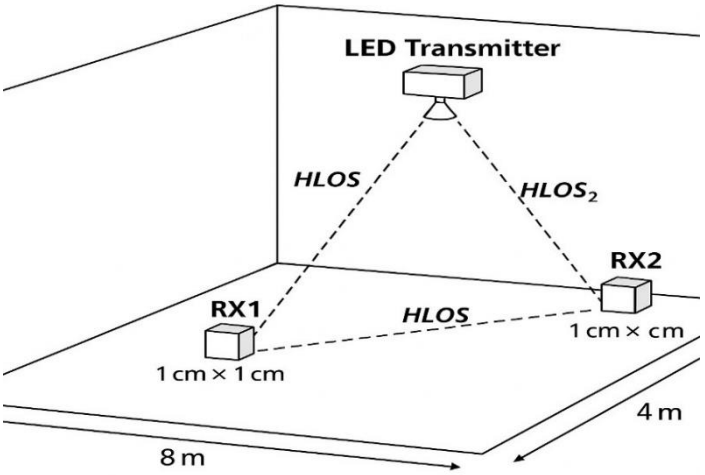


Fig 2.2: Enhanced Image Transmission

2.3 PERFORMANCE PARAMETERS

Since the image data is used for communication a comparison between cooperative and direct transmission is made using the parameters listed below.

2.3.1 Mean Square Error (MSE)

A popular statistic for assessing the authenticator quality of picture reception is MSE. It calculates the mean squared difference between the processed or received image and the original image. The following formula can be used to determine the

$$MSE = (1 / (k \times l)) \times \Sigma\Sigma [I(m, n) - \hat{I}(m, n)]^2 \quad \ldots.(4)$$

where, k and l are the dimensions (width and height) of the images. I(m, n) and ˆI(m, n) represents the intensity (pixel value) of the original and reconstructed image respectively, at position (m, n).

2.3.2 Peak Signal to Noise Ratio (PSNR)

In image processing, the PSNR is a frequently used metric to assess how closely a picture has been processed to its original state. The PSNR is defined as

$$PSNR = 10 \times \log_{10} (PS^2 / MSE) \quad \ldots.(5)$$

where, P S is the peak signal value.

2.3.3 Normalized Cross Correlation (NCC)

A similarity metric called NCC is used in image processing to determine how similar two images are to one another. It measures the degree to which one image region resembles another. The following is the NCC formula

$$NCC = \Sigma [(I(m, n) - \mu_i) \times (\hat{I}(m, n) - \mu\hat{i})] / (\sigma_i \times \sigma\hat{i}) \quad \ldots..(6)$$

where, μI and μˆI are the means, while σI and σˆI are the standard deviations of the pixel intensities of images I and ˆI, respectively.

2.3.4 Normalized Absolute Error (NAE)

In image processing, the NAE metric is used to measure the inaccuracy between two pictures. After normalizing the outcome using the dynamic range of the pixel values, NAE calculates the absolute inaccuracy between corresponding pixel values in the original and processed photos. The following is the formula for the NAE

$$NAE = (1 / (k \times l \times t)) \times \Sigma\Sigma [I(m, n) - \hat{I}(m, n)]^2 \quad \ldots..(7)$$

where, the dynamic range of pixel values, t, often equates to the highest conceivable pixel value, which is established by the image's bit depth (e.g., 255 for an 8-bit grayscale image).

2.3.5 Laplacian Mean Squared Error (LMSE)

An image quality metric called LMSE is used in image processing to measure the difference or error between two images. It is applied after Laplacian sharpening has improved the images, and it is based on the MSE. The following is the formula for the LMSE

$$LMSE = (1 / (k \times l)) \times \Sigma\Sigma [L(I(m, n)) - L(\hat{I}(m, n))]^2 \quad \ldots.(8)$$

where, L(I(m, n)) and L(ˆI(m, n)) represents the Laplacian-enhanced intensity of the original and received image respectively.

3. EXPERIMENTS AND RESULTS

3.1 Performance Parameters Plots :

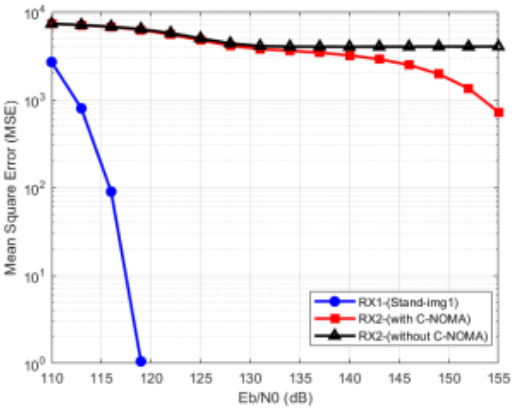


FIG.3.1(a) MSE VS SNR

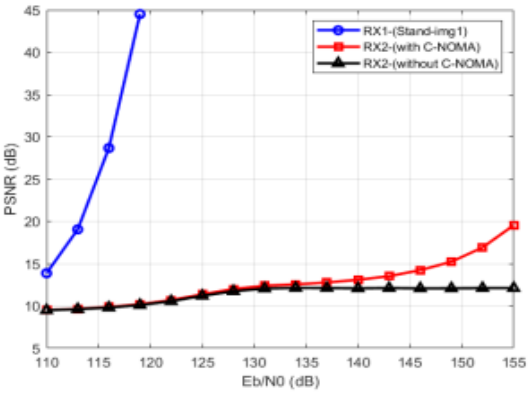


FIG.3.1(b) PSNR VS SNR

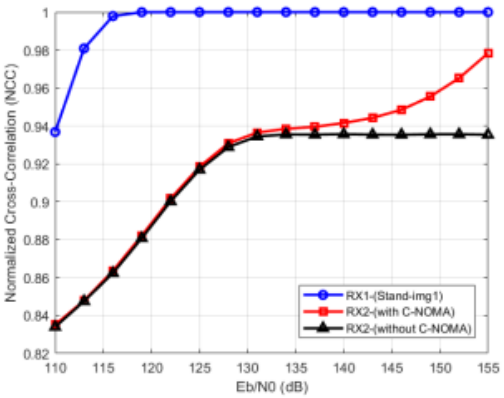


FIG.3.1(c) NCC VS SNR

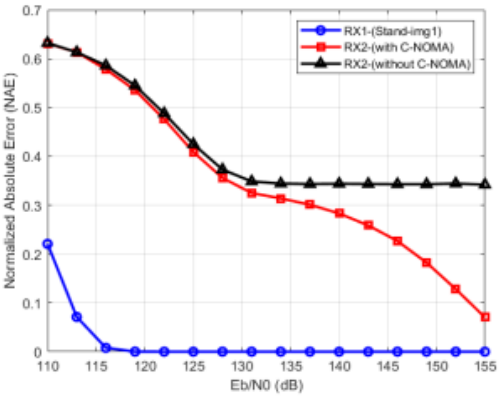


FIG.3.1(d) NAE VS SNR

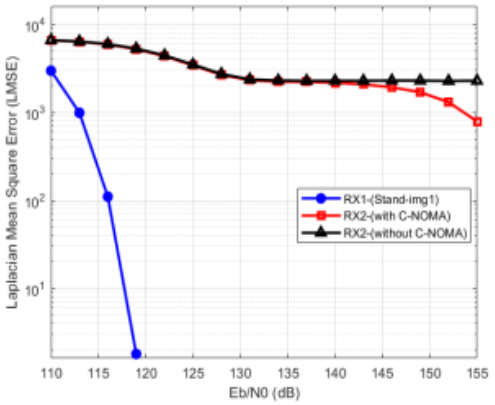


FIG.3.1(e) LMSE VS SNR

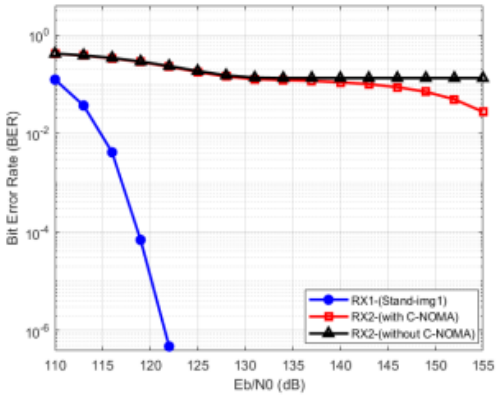


FIG.3.1(f) BER VS SNR

3.2 OUTPUT IMAGES :



fig.3.2(a) Input image1



fig.3.2(b) Input image2



fig.3.2(c) Output image1



fig.3.2(d) Output image2

3.3 PERFORMANCE PARAMETERS :

P1	MSE		PSNR (dB)		NCC		NAE		LMSE	
	Rx1	Rx2	Rx1	Rx2	Rx1	Rx2	Rx1	Rx2	Rx1	Rx2
0.5	9328.74	9328.74	8.4300	8.4300	0.8133	0.9204	0.5891	0.6871	2.0290	72.0951
0.6	9328.74	9328.74	204.6700	8.4300	0.8133	0.9204	0.5891	0.6871	2.0290	72.0951
0.7	0.00	9328.74	204.6700	8.4300	1.0000	0.9204	0.0000	0.6871	0.0000	72.0951
0.8	0.00	83.27	204.6770	28.9300	1.0000	0.9970	0.0000	0.0085	0.0000	10.8892
0.9	0.00	0.0000	204.6700	204.6700	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000

4. APPLICATIONS

The integration of Cooperative NOMA in VLC networks significantly improves the performance of image data transmission, especially in environments with complex indoor conditions and high user density. This section explores key application areas that benefit from enhanced image delivery capabilities.

4.1. IMAGE TRANSMISSION

- Visible Light Communication (VLC) provides a promising solution for high speed, secure image data transmission, especially in RF-restricted environments. Cooperative relaying in VLC networks ensures that even users outside the direct line-of-sight (LoS) can reliably receive image data, thus enhancing coverage and reliability.
- Use Case: Real-time transmission of high-resolution images in indoor settings such as museums, retail stores, or galleries.
- Benefit: Reduced latency and increased reliability in delivering large image files, especially in scenarios with dynamic obstacles or poor channel conditions.
- Advantage of Cooperative NOMA: Simultaneous transmission to multiple users with power-domain multiplexing, while ensuring weaker users benefit from relayed information, minimizing transmission errors.

4.2. SMART INDOOR SURVEILLANCE SYSTEMS

In smart surveillance, cameras continuously capture and transmit image and video data to a monitoring station. VLC is ideal for indoor environments where RF interference or security concerns make traditional wireless communication less effective.

- Use Case: A VLC - enabled surveillance system in a shopping mall or office transmits high-definition snapshots of crowd movement or object recognition data to a central security server.
- Challenge: Obstructions, such as walls or partitions, reduce LoS efficiency.
- Solution: Cooperative users (relays) receive data and forward it to users outside the LoS, ensuring complete area coverage.
- Advantage: Enhances the integrity and speed of transmitted image data, reduces black spots in camera networks, and increases resilience to signal blockages.

4.3. INDUSTRIAL AUTOMATION AND MONITORING

- Use Case: Transmission of visual inspection images from conveyor belts to control units in a manufacturing plant.
- Challenge: Moving equipment or structural barriers may block direct communication paths.
- Solution: Cooperative VLC networks allow mobile or fixed relays to extend connectivity, ensuring uninterrupted image flow.
- Benefit: Low-latency, high-reliability image delivery helps in detecting faults early, reducing downtime and improving automation efficiency.

4.4. HEALTHCARE AND MEDICAL IMAGING

- Use Case: Transmitting medical scans from imaging equipment in radiology departments to doctors' workstations across different rooms or floors.
- Challenge: VLC line-of-sight limitations and the need for fast, uninterrupted data flow.
- Solution: Cooperative user nodes act as trusted relays to forward image data while maintaining privacy and speed.
- Advantage: Higher data security (as VLC signals don’t penetrate walls), minimal latency, and consistent image quality even under variable channel conditions.

5. CONCLUSION

The performance of image transmission over a co-operative VLC-NOMA system has been analyzed. The system utilizes OOK modulation along with decode-and forward (DF) relaying and maximum ratio combining (MRC) to improve communication reliability. The performance evaluation is carried out using different image quality metrics such as MSE, PSNR, NCC, NAE, LMSE, and BER. The simulation results show that the co operative transmission scheme significantly outperforms the non-cooperative approach. The MSE is reduced, PSNR is increased, and NCC values are closer to unity, indicating better image reconstruction quality. Furthermore, the BER performance also improves with co-operative transmission due to the presence of relay-assisted communication, which enhances the effective SNR and re duces transmission errors. Fig. 10. Input image2 used for VLC transmission Fig. 11. Image Transmission without Cooperation Fig. 12. Image Transmission with Cooperation Overall, the results demonstrate that cooperative communication plays a crucial role in improving the reliability, efficiency, and quality of image transmission in VLC systems, especially for users located far from the transmitter. Future work may include the implementation of advanced modulation techniques, adaptive power allocation, and deep learning-based decoding methods to further enhance system performance.

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7. PUBLICATRION

- Pravind Pal, Anish Verma, Daniel Somabattula, Aman Kumar Barnwal, Atul Kumar, Gaurav Upadhyay " On Image Transmission Using Cooperative NOMA in VLC Network ," IETE Journal of Research, to be submitted.

On Image Transmission Using Cooperative NOMA In VLC Network

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***Abstract*—Visible light communication uses LED lights to send information without using cables. The LED lights get brighter and darker fast, thousands of times every second. This change in brightness happens so fast that we cannot see it, but a special receiver can detect it and figure out what it means. Visible light communication works with the kind of light that we can see, which is between 380 and 780 nanometers. One good thing about light communication is that a lot of people can use the same light at the same time. There is a way to make this work called "non-orthogonal multiple access." Non-Orthogonal Multiple Access helps people who are away from the light or have a weak signal get the information they need. This means that the whole system can handle information.**

When we work together with light communication, it becomes more reliable. This helps the signal go farther and work better when things are in the way. For example, if something is blocking the light from getting to the receiver, we can still send the information. This is really helpful for people who're far away from the hospital.

In our study we sent picture information using light communication. We used a way of sending information called "on-off keying." With on-off keying, the light is either all the way on or all the way off to represent the information. Finally, we compared how well picture information is sent when we work together with light communication and when we do not, using both on-off keying and another way called maximum ratio combining. We wanted to see which way works better for sending picture information with light communication.

***Index Terms*—Index Terms—On-Off Keying (OOK), Mean Square Error (MSE), Peak Signal to Noise Ratio (PSNR), Normalized Cross Correlation (NCC), Normalized Absolute Error (NAE), Laplacian Mean Squared Error (LMSE) Demand for higher data rates has always been one of the main challenges in telecommunications.**

I. Introduction

People are using more electronics every year, and at the same time, new apps and other online services are being introduced. Due to this growth of internet traffic [1], the overall amount of wireless signals is increasing exponentially. This growth creates a problem where many wireless signals interfere with each other. Therefore, networks will reuse a common radio frequency repeatedly, and doing so only creates additional interference; consequently, there is not enough radio frequency (RF) spectrum available for all users. Also, in addition to the interference from many wireless signals, there are many limitations of the electromagnetic spectrum itself, which are limited capacity, weak security, low availability, and low efficiency.

An example of a technology that can help with these challenges is called visible light communication (VLC). VLC is a form of wireless communication that utilizes visible light from common light sources like light-emitting diodes (LEDs) or incandescent bulbs to transmit data wirelessly. The light source is modulated (flickered) at very high speeds, and the human eye cannot detect this flickering. The flickering light is received by a special photodetector that converts it back into the original data in its unmodified form. Because light is abundant in almost all indoor spaces, VLC provides a very low-cost, secure solution for transmitting data wireless without contributing to current levels of wireless radio frequency congestion.

A very useful idea for sending pictures using light. Pictures have been one of the most popular methods that people use today to share information. For example, at business meetings and in teaching classrooms using photographs or during storytelling and creating digital art, photographs have become an important way for us all to communicate. An On-Off Keying (OOK) modulation method is used in VLC systems. OOK MODULATION uses a flash of light to represent a binary "1" and no light to represent a binary "0." This is a straightforward technique for converting digital data into light signals and from light signals back to digital data.

Enhancing cooperative communication will improve wireless network communications' capacity, speed, and overall performance. Cooperative communication will also extend the communication distance of both mobile networks and ad-hoc networks. Decode-and-forward relay (DFR) communication will be used as the principal technique in this article. In DFR communication, a middle user assists in relaying the signal, or message, to the final destination. The middle user has three functions in this process: receiving the received signal, decoding the received signal, and re-sending the re-encoded signal to the final destination. Thus, with DFR communication, the reliability of communication in VLC systems will be enhanced, particularly when the end user is located a great distance from the original transmitter and/or when the light has been obstructed by walls, furniture, and other objects.

Research published until now on cooperative VLC systems has focused on cooperative VLC systems in general. Nevertheless, only two of the wide range of papers published to date have investigated how cooperative VLC systems work for sending images. One paper discusses a hybrid approach where

RF is used with VLC to provide a mechanism for cooperative communications between vehicles. The other discusses a cooperative VLC system that uses On-Off Keying (OOK) modulation in both ideal and non-ideal channel conditions. The other two papers are as follows: One proposes a relaying strategy whereby the relay receives the image data via VLC from the sender and transmits the image data to the destination via RF; and the other presents a NOMA (Non-Orthogonal Multiple Access)-based cooperative VLC system that combines OOK and L-Pulse Position Modulation to relay images to the destination. Researchers have also examined MIMO-VLC systems with receivers oriented at various angles, as well as the impact of energy consumption and alternative modulations on the potential for signal outage in systems employing selective/incremental Decode and Forward (DF) relaying systems.

The authors of this paper contend that there has not been any previous research that compares cooperative (i.e., cooperatively relaying the image data) and non-cooperative (i.e., sending the image data directly) VLC systems using OOK modulation and Maximum Rank Communication (MRC), and thus, this paper fills that research gap.

The layout of the final portion of this document is kept simple. Section 2 goes into detail about the OOK based VLC system as well as system and channel models, and the packet structure for signals. Section 3 discusses performance evaluations used in comparison of non-cooperative versus cooperative communication, including metrics such as MSE and PSNR, etc. Section 4 presents experimental results with discussion. Section 5 concludes with a recap of the findings and provides suggestions for future research.

II. COLLABORATIVE DATA TRANSMISSION IN VLC NETWORKS

A. System Configuration and Geometric Arrangement

In an indoor visible light communication system, we have one node that sends information and two nodes that receive it. One of these receiving nodes helps the other one get the information and the other one is the destination. The node that sends information has one emitting diode or LED and it is placed in the middle of the ceiling of a room that is 8 meters long 8 meters wide and 4 meters high. The light from this LED goes out in a pattern and this pattern is important for the nodes that receive the information.

The node that helps the one, called U1 has two main parts. One part receives the information. It has a special device called a photodetector or PD. This device looks at the light. Figures out what the message says. Then it sends the message to the part of U1, which sends it out again as light. This light has a pattern and it is strong enough to reach the final destination.

When we use Light Communication in a big room it is helpful to have nodes that can help each other. This is because the light can only go far and it can be blocked by things in the room. So we use a system where one node helps another node get the information. We can see a picture of this system in Figure 1 where two nodes, U1 and U2 work together to get the information.

When we send information there are two steps. The first step is called the broadcast step. In this step the node that sends information called Tx sends the message to all the nodes. The second step is called the cooperation step. In this step U1 helps U2 get the message. U1 sends the message again. U2 gets two messages, one from Tx and one, from U1. Then U2 uses a technique to combine these two messages and get the information it needs.

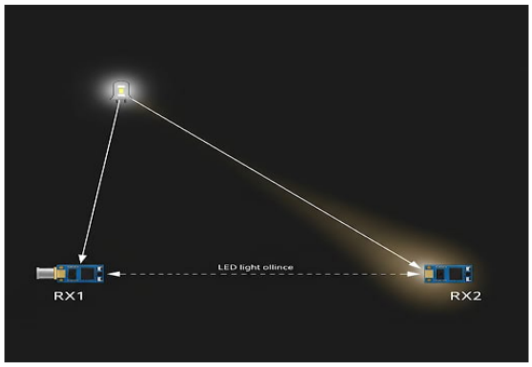


Fig. 1. System Model for 2 User Cooperative VLC System

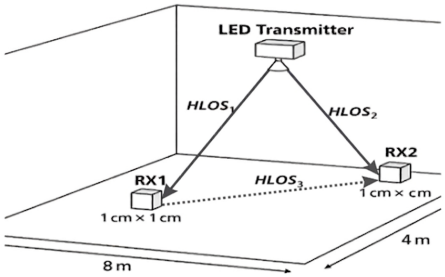


Fig. 2. Room Setup for VLC System

The line-of-sight (LOS) channel impulse response (CIR) is given by[14]

$$H_{LOS} = \begin{cases} \frac{A}{D_{i,j}^2} R_\ell(\psi_i)\, G_c(\phi_{i,j})\, G_s(\phi_{i,j}) \cos(\phi_{i,j}), & 0 \le \phi \le \Phi_c \\ 0, & \phi > \Phi_c \end{cases} \tag{1}$$

where A represents the physical area of the photodetector, $D_{i,j}$ represents the Euclidean distance between the ith transmitter and j^{th} receiver, ϕ represents the angle of incidence, Φ_c represents the field of view (FoV) of Pphto-Detector, and ψ represents the irradiance angle. $G_c(\phi_{i,j})$ represents the gain of optical filter,

The gain of an optical concentrator is given by:

$$G_s(\phi_{i,j}) = \frac{n^2}{\sin^2(\Phi_c)}, \quad 0 \le \phi \le \Phi_c \tag{2}$$

Where n is the refractive index of the optical concentrator. The Lambertian radiant intensity is given by:

$$R_\ell(\psi_i) = \frac{m+1}{2\pi}\cos^n(\psi_i) \tag{3}$$

where n represents the order of Lambertian emission defined as:

$$m = \frac{-\ln(2)}{\ln(\cos(\psi_{1/2}))} \tag{4}$$

represents the order of Lambertian emission represented by semi-angle at half power of an LED [15].

B. Signal Structure

The transmitted signal can be expressed as $s = 2P_t b$, where b represents the transmitted symbol and P_t denotes the average transmitted power. The signal received at the kth receiver from the lth transmitter is given by

$$r_k = 2\rho H_{LOS} P_t b + w \tag{5}$$

where H_{LOS} represents the LOS channel gain between the lth transmitter and k^{th} receiver, ρ is the responsivity of the photodetector, and w denotes additive white Gaussian noise (AWGN).

The noise variance is given by

$$\sigma_w^2 = \sigma_{th}^2 + \sigma_{sh}^2 \tag{6}$$

where σ_{th}^2 and σ_{sh}^2 compares the received signal against a threshold, and then uses the decode-and-forward (DF) method to relay the detected signal to the destination, D_1.

The destination will receive both the signal sent directly by the source and the signal sent by the relay. The two signals will be combined using maximum ratio combining (MRC) prior to being combined.

Prior to combining the two signals, each signal will be assigned a weighting factor based on the quality of the signal. This will result in an improved overall signal-to-noise ratio (SNR) and an improved quality of reception [8].

TABLE I
SIMULATION PARAMETERS

Description	Value
Size of Room	$8\,m \times 8\,m \times 4\,m$
Source Coordinates	$4\,m \times 4\,m \times 4\,m$
U_1 Coordinates	$7\,m \times 6\,m \times 1.5\,m$
U_2 Coordinates	$4\,m \times 4\,m \times 1.5\,m$
Semi-half angle of LED ($\psi_{1/2}$)	60°
FoV (Φ_c)	60°
Physical Area of PD (A)	$1\,\text{cm}^2$
Refractive index (n)	1.5
Gain of optical filter $G_c(\phi_{i,j})$	1
Transmission rate (R_b)	50 Mbps
Noise bandwidth (B)	50 MHz
Background Current (I_B)	$202\,\mu A$

III. PERFORMANCE PARAMETERS

The performance of the proposed cooperative VLC-NOMA system is evaluated using multiple image quality and error metrics, including MSE, PSNR, NCC, NAE, LMSE, and BER. These parameters are analyzed with respect to varying SNR values to understand the effectiveness of cooperative transmission.

A. Mean Square Error (MSE)

MSE allows you to determine how closely an original image matches its reconstructed form based on the average squared distance between the differing pixels of the images. The lower the MSE value is, the more accurate the reconstructed image will be to the original image with minimal visual errors/distortions.

$$MSE = \frac{1}{P \times Q}\sum_{p=1}^{P}\sum_{q=1}^{Q}\left[I(p,q) - \hat{I}(p,q)\right]^2 \tag{7}$$

where, P and Q are the dimensions (width and height) of the images. $I(p,q)$ and $\hat{I}(p,q)$ represent the pixel intensity values of the original and reconstructed images, respectively, at position (p,q).

B. Peak Signal to Noise Ratio (PSNR))

PSNR is a metric used to compare a reconstructed image against an original image to determine quality. Like MSE, PSNR is expressed in decibels (dB) and functions oppositely of MSE; when MSE values decrease (indicating less distortion), corresponding PSNR values will increase (indicating an improved match). Thus, a higher PSNR indicates fewer defects in the reconstructed image.

$$PSNR = 10\log_{10}\left(\frac{I_{max}^2}{MSE}\right) \tag{8}$$

where, I_{max} represents the maximum possible pixel value (e.g., 255 for 8-bit images).

C. Normalized Cross Correlation (NCC)

NCC is a method used to compare two images by comparing the relative alignment of their corresponding pixel patterns. The NCC score ranges between -1 and 1; thus, the higher the NCC score is (i.e., the closer the NCC score is to 1) indicates the reconstruction's legs are more similar to those of the original image.

$$NCC = \frac{\sum_{p=1}^{P}\sum_{q=1}^{Q}(I(p,q) - \bar{I})(\hat{I}(p,q) - \bar{\hat{I}})}{\sigma_I \cdot \sigma_{\hat{I}}} \tag{9}$$

where, $\bar{I}$ and $\bar{\hat{I}}$ are the mean pixel values, while σ_I and $\sigma_{\hat{I}}$ are the standard deviations of the original and reconstructed images, respectively.

D. Normalized Absolute Error (NAE)

Normalized Absolute Error (NAE) examines the reconstruction accuracy of an image by dividing the total absolute differences (pixel by pixel) between the identical pixel values and the lowest possible pixel value by the percentage difference, using the entire possible range of pixel values as the denominator. This ensures that the amount of overall brightness/darkness does not affect the NAE calculation, resulting in a smaller NAE for improved reconstruction accuracy and decreased distortion.

$$NAE = \frac{1}{P \times Q \times L} \sum_{p=1}^{P} \sum_{q=1}^{Q} \left| I(p,q) - \hat{I}(p,q) \right|^2 \quad (10)$$

where, P and Q represent the dimensions of the image, L denotes the maximum possible pixel value (e.g., 255 for 8-bit images), and $I(p,q)$ and $\hat{I}(p,q)$ represent the original and reconstructed pixel values at position (p,q).

E. Laplacian Mean Squared Error (LMSE)

The LMSE metric compares the quality of an image by first applying a Laplacian filter to both original and reconstructed images to detect edge and fine-detail increases. The LMSE metric then compares the two Laplacian filtered images and reports an LMSE value. The lower the LMSE value is, the more successful the reconstructed image will be in preserving edge and small-detail information.

$$ALMSE = \frac{1}{P \times Q} \sum_{p=1}^{P} \sum_{q=1}^{Q} \left[L(I(p,q)) - L(\hat{I}(p,q)) \right]^2 \quad (11)$$

where, $L(I(p,q))$ and $L(\hat{I}(p,q))$ represent the Laplacian-transformed intensities of the original and reconstructed images, respectively.

IV. RESULT AND ANALYSIS

The performance of the proposed VLC system is evaluated under both cooperative and non-cooperative transmission schemes. The analysis is carried out over a range of Signal-to-Noise Ratio (SNR) values to examine the behavior of different performance metrics such as MSE, PSNR, NCC, NAE, and LMSE.

Fig. 3 illustrates the variation of Mean Square Error (MSE) with respect to SNR. It is observed that the MSE decreases as the SNR increases, indicating improved reconstruction of the transmitted image. Moreover, the cooperative system consistently exhibits lower MSE compared to the non-cooperative system, which confirms reduced distortion and better accuracy.

Fig. 4 shows the PSNR performance. It can be seen that PSNR increases with increasing SNR, implying better image quality at higher signal strengths. The cooperative approach achieves higher PSNR values than the non-cooperative case, demonstrating enhanced visual quality of the received image.

The variation of Normalized Cross-Correlation (NCC) is depicted in Fig. 5 The NCC values move closer to unity as SNR increases, indicating higher similarity between the original and reconstructed images. The cooperative transmission yields higher NCC values at all SNR levels, confirming better structural similarity.

Fig. 6 presents the Normalized Absolute Error (NAE) analysis. The NAE decreases with increasing SNR, showing a reduction in pixel-wise error. It is evident that the cooperative system produces lower NAE values compared to the non-cooperative system, indicating improved accuracy in image reconstruction.

Fig. 7 shows the LMSE performance, which focuses on edge preservation and detail accuracy. The LMSE decreases as SNR increases, and the cooperative system consistently outperforms the non-cooperative system by producing lower LMSE values.

Fig. 8 shows that the Bit Error Rate (BER) decreases as the SNR increases, indicating improved communication reliability. The cooperative system achieves lower BER than the non-cooperative system due to relay-assisted transmission and MRC combining, resulting in better error performance.

V. IMAGE ANALYSIS

To visually demonstrate the improvement achieved by incor-porating the cooperative NOMA, this section presents the reconstructed images at the receiver for four cases: the first original image used for transmission,the Second original image used for transmission, the image received at user Rx1 , the image received at user Rx2. The comparison clearly highlights the benefits of indicating better structural similarity , improved reconstruction of the transmitted image.

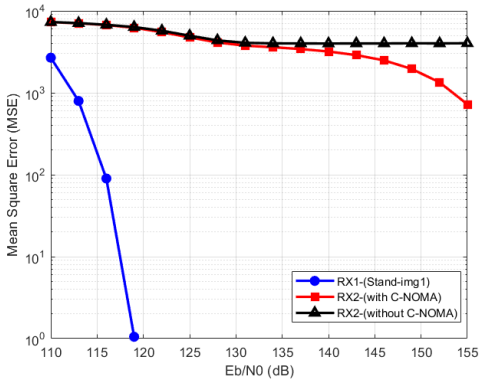


Fig. 3. MSE VS SNR

TABLE II
PERFORMANCE ANALYSIS OF VLC SYSTEM

P1	MSE		PSNR (dB)		NCC		NAE		LMSE	
	Rx1	Rx2	Rx1	Rx2	Rx1	Rx2	Rx1	Rx2	Rx1	Rx2
0.5	9328.74	9328.74	8.43	8.43	0.8133	0.9204	0.5891	0.6871	2.0290	72.0951
0.6	9328.74	9328.74	8.43	8.43	0.8133	0.9204	0.5891	0.6871	2.0290	72.0951
0.7	0.0000	9328.74	204.67	8.43	1.0000	0.9204	0.0000	0.6871	0.0000	72.0951
0.8	0.0000	83.2715	204.67	28.93	1.0000	0.9970	0.0000	0.0085	0.0000	10.8892
0.9	0.0000	0.0000	204.67	204.67	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000

VI. CONCLUSION

In this paper, the performance of image transmission over a cooperative VLC-NOMA system has been analyzed. The

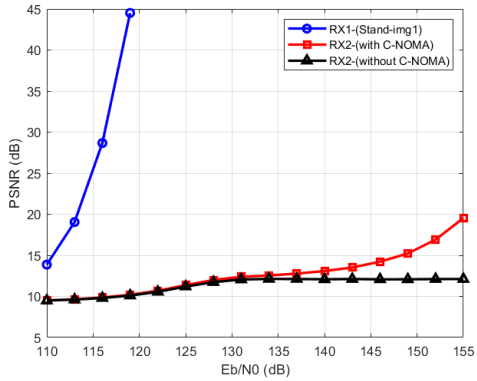


Fig. 4. PSNR VS SNR

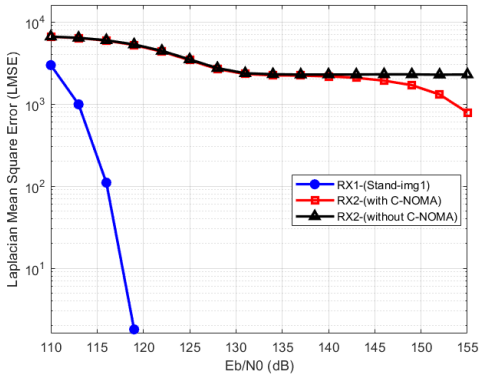


Fig. 7. LMSE VS SNR

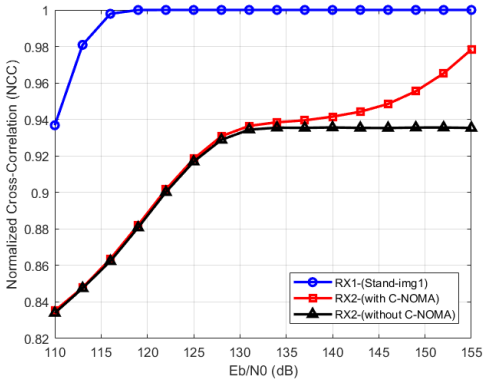


Fig. 5. NCC VS SNR

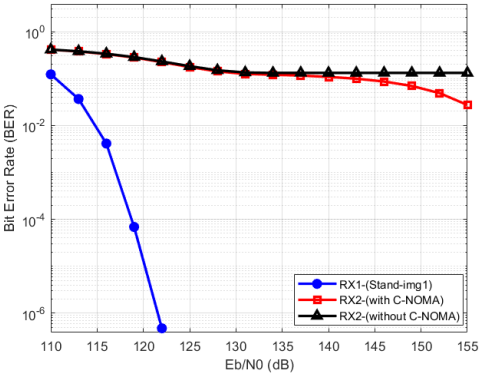


Fig. 8. BER VS SNR

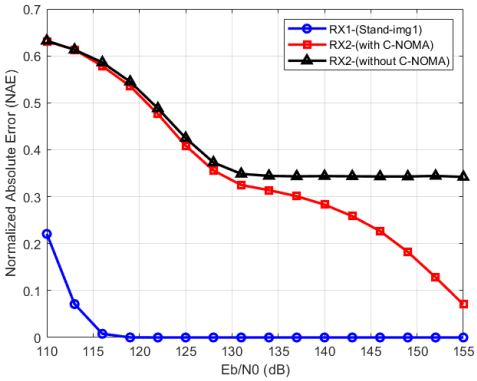


Fig. 6. NAE VS SNR



Fig. 9. Input image1 used for VLC transmission



Fig. 10. Input image2 used for VLC transmission



Fig. 11. Output Image at RX1



Fig. 12. Output Image at RX2

system utilizes OOK modulation along with decode-and-forward (DF) relaying and maximum ratio combining (MRC) to improve communication reliability.

The performance evaluation is carried out using different image quality metrics such as MSE, PSNR, NCC, NAE, LMSE, and BER. The simulation results show that the cooperative transmission scheme significantly outperforms the non-cooperative approach. The MSE is reduced, PSNR is increased, and NCC values are closer to unity, indicating better image reconstruction quality.

Furthermore, the BER performance also improves with cooperative transmission due to the presence of relay-assisted communication, which enhances the effective SNR and reduces transmission errors.

Overall, the results demonstrate that cooperative communication plays a crucial role in improving the reliability, efficiency, and quality of image transmission in VLC systems, especially for users located far from the transmitter.

Future work may include the implementation of advanced modulation techniques, adaptive power allocation, and deep learning-based decoding methods to further enhance system performance.

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